King Saud University
College of Business Administration
Management Information Systems Department

MIS 214- Introduction to Databases
1nd Semester 2025

Ithra Library Book Borrowing System
Database Design Project

Database system designed to manage book borrowing, reservations, and fines in a library environment.

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Project Description: Ithra Library Book Borrowing System

This project focuses on designing and developing a database application for the Book Borrowing Department at Ithra Library. The client is the Ithra Center, which aims to provide a seamless borrowing experience for its visitors.

The system will manage information about books, members, borrowing transactions, and staff. It will allow librarians to record and track which books are borrowed, by whom, and when they are due for return. The system will also include features for managing book availability, member registration, and overdue notifications.

The goal of this database is to organize the library's borrowing operations, and provide quick access to accurate information. This will improve efficiency, prevent data loss, and enhance the overall user experience for both staff and visitors.

View Description

The BorrowedBooksView is created to display all active borrowing transactions in the Ithra Library database. It shows important details such as the book title, member name, borrowing date, return due date, and the staff member who processed the borrowing.

This view will mainly be used by librarians to check which books are currently borrowed, who borrowed them, and which ones are overdue. It helps them manage book returns and send reminders more efficiently.

The view allows users to access borrowing information directly without searching through multiple tables, making their work faster, easier, and more organized.

Data Requirements

Book Entity:

The Book entity stores information about all books available in the library. Each book is uniquely identified by **Book_ID**, and includes attributes such as Title, Author, ISBN, Genre, Publication_Year, Status, and Shelf_Location. The Book entity has a one-to-many relationship with both the Borrow and Reservation entities.each book can be borrowed or reserved by multiple members. This entity is involved in the borrow and reservation transactions, where the system updates a book’s status to indicate whether it is available, borrowed, or reserved.

Member Entity:

The Member entity represents individuals who are registered to use the library’s services. Each member is identified by a unique **Member_ID**, and includes attributes such as Name, Phone, Email, Membership_Date, Expiry_Date, Max_Book_Allowed, and Status. A single member may borrow or reserve several books, the member entity has a one-to-many relationships with both the Borrow and Reservation entities. This entity participates in borrow, return, and reservation transactions, as each member can borrow, return, or reserve books through the system.

Employee Entity:

The Employee entity contains data about the library’s staff who manages operations. Each employee has a unique **Employee_ID**, and attributes include Name, Position, Phone, and Email. Each employee may handle multiple borrow transactions, but each transaction is managed by only one employee, there is a one-to-many relationship between Employee and Borrow. The relationship between the Employee and reservation entity is optional as members can make reservations through the system without the staff involvement unless they are asked to.This entity is directly involved in the borrow transaction, where employees record and authorize the lending process for books.

Borrow Entity:

The Borrow entity records details of all borrowing activities in the library. Each transaction is uniquely identified by **Borrow_ID**, and includes Book_ID (FK), Member_ID (FK), Employee_ID (FK), Borrow_Date, Actual_Return_Date, Due_Date, Status, and Fine. It establishes relationships among Books, Member, and Employee, representing which member borrowed which book and which employee processed the transaction. This entity is central to borrow and return transactions, as it tracks borrowing dates, due dates, and fines for overdue books.

Reservation Entity:

The Reservation entity stores data about books reserved by members. It includes unique **Reservation_ID**, Book_ID (FK), Member_ID (FK), Reservation_Date, Status and Expiry_Date. Each reservation connects one book with one member, where a member can make several reservations and a book can be reserved by multiple members. This entity is involved in reservation transactions, which record when a member reserves a specific book, set the reservation’s expiry date, and update the book’s status.

Transaction Requirements:

Data Entry

1-Enter the details of a new book

2-Enter the details of a new member

Data Update/Deletion

1-Update/delete membership details

2-Update/delete book shelf location

3-Update/delete contact information for a member

Data Queries

1-List all books currently borrowed by a specific member.

2-List the top 3 most borrowed books.

3-List reservations made within the last 7 days.

4-List members who haven't returned their borrowed books yet.

5-List all books that have never been borrowed.

6-List the top 3 most borrowed genres.

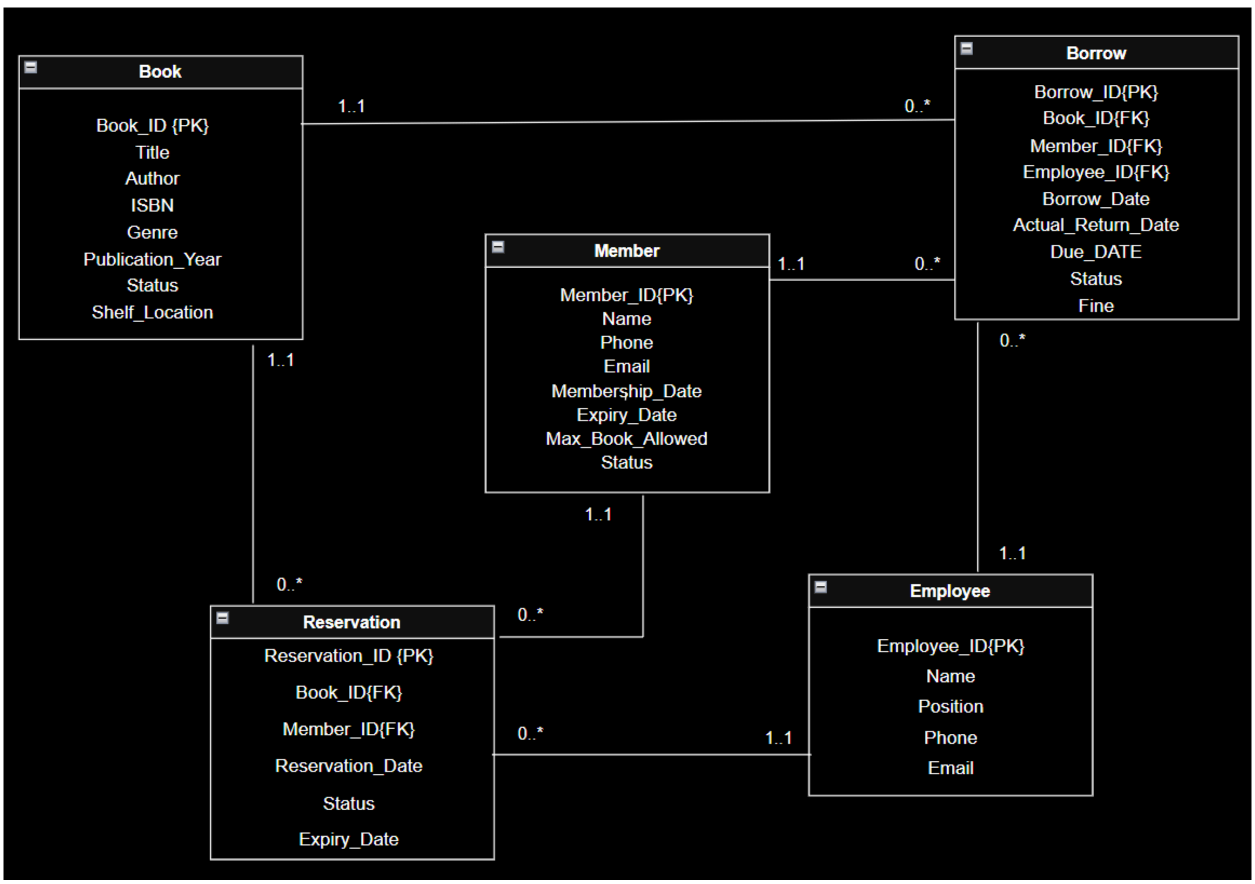
7-List which employee processed the most borrowings.

8-Find the sum of all fines collected on a specific date.

9-List members who have reached their maximum allowed book limit.

10-List members who returned books late.

Enhanced Entity Relationship Diagram (EER)



The EER diagram was mapped into a relational schema with clearly defined primary and foreign keys to enforce referential integrity

Book(Book_ID PK, Title, Author, ISBN, Genre, Publication_Year, Status, Shelf_Location)

Member(Member_ID PK, Name, Phone, Email, Membership_Date, Expiry_Date, Max_Book_Allowed, Status)

Borrow(Borrow_ID PK, Book_ID, Member_ID ,Employee_ID FK, Borrow_Date, Actual_Return_Date,Due_DATE, Status, Fine)
FK Book_ID references Book (Book_ID)
Member_ID references Member (Member_ID)
Employee_ID references Employee (Employee_ID)

Employee(Employee_ID PK, Name, Position, Phone, Email)

Reservation(Reservation_ID PK, Book_ID, Member_ID, Employee_ID, Reservation_Date, Status, Expiry_Date)
FK Book_ID references Book(Book_ID)
Member_ID reference Member(Member_ID)
Employee_ID references Employee (Employee_ID)

2.Write the SQL DDL statements for creating the tables based on your ER diagram.

In this section, SQL Data Definition Language (DDL) statements are written to create the database tables based on the Enhanced Entity Relationship (EER) diagram of the Ithra Library Book Borrowing System. Each table represents an entity from the diagram, with properly defined primary keys, foreign keys, data types, and necessary constraints such as NOT NULL, UNIQUE, and DEFAULT. The relationships among the entities are established using foreign keys to ensure referential integrity within the database.CREATE DATABASE LibraryDB;

```
USE LibraryDB;


CREATE TABLE Employee(

   Employee_ID     char(20) PRIMARY KEY NOT NULL,

   Name           varchar(40) NOT NULL,

   Position       varchar(40),
```

```
    Phone          varchar(20),
    Email          varchar(40)
);

CREATE TABLE Member(
    Member_ID      char(20) PRIMARY KEY NOT NULL,
    Name           varchar(40) NOT NULL,
    Phone          varchar(20),
    Email          varchar(40),
    Membership_Date date,
    Expiry_Date    date,
    Max_Book_Allowed numeric(2,0),
    Status         varchar(20)
);

CREATE TABLE Book(
    Book_ID        char(20) PRIMARY KEY NOT NULL,
    Title          varchar(80) NOT NULL,
    Author         varchar(40),
    ISBN           varchar(30),
    Genre          varchar(30),
    Publication_Year numeric(4,0),
    Status         varchar(20),
    Shelf_Location  varchar(40)
);

CREATE TABLE Borrow(
    Borrow_ID          char(20) PRIMARY KEY NOT NULL,
```

```
    Book_ID          char(20) NOT NULL,
    Member_ID        char(20) NOT NULL,
    Employee_ID      char(20) NOT NULL,
    Borrow_Date      date NOT NULL,
    Actual_Return_Date  date,
    Due_Date         date,
    Status           varchar(20),
    Fine             numeric(8,2),

    FOREIGN KEY (Book_ID) REFERENCES Book(Book_ID)
      ON DELETE CASCADE ON UPDATE CASCADE,

    FOREIGN KEY (Member_ID) REFERENCES Member(Member_ID)
      ON DELETE CASCADE ON UPDATE CASCADE,

    FOREIGN KEY (Employee_ID) REFERENCES Employee(Employee_ID)
      ON DELETE CASCADE ON UPDATE CASCADE
);

CREATE TABLE Reservation(
    Reservation_ID   char(20) PRIMARY KEY NOT NULL,
    Book_ID          char(20) NOT NULL,
    Member_ID        char(20) NOT NULL,
    Employee_ID      char(20) NOT NULL,
    Reservation_Date date NOT NULL,
    Expiry_Date      date,
    Status           varchar(20),
```

```
   FOREIGN KEY (Book_ID) REFERENCES Book(Book_ID)
      ON DELETE CASCADE ON UPDATE CASCADE,

   FOREIGN KEY (Member_ID) REFERENCES Member(Member_ID)
      ON DELETE CASCADE ON UPDATE CASCADE,

   FOREIGN KEY (Employee_ID) REFERENCES Employee(Employee_ID)
      ON DELETE CASCADE ON UPDATE CASCADE
);
```

Each table is carefully structured to reflect the entities and relationships in the system:
- The Book table stores details about available books in the library.
- The Member table manages information about library members and their borrowing privileges.
- The Employee table stores staff details responsible for processing transactions.
- The Borrow table records all borrowing transactions and links books, members, and employees.
- The Reservation table tracks book reservations made by members.

This structure ensures data consistency, supports efficient borrowing operations, and maintains accurate tracking of books and members within the Ithra Library system.

3. Write the DML statements for your database

Data Entry:

1-INSERT INTO Book VALUES

('B01', 'Pride and Prejudice', 'Jane Austen', '9780735211292', 'Fiction', 1813, 'Available', 'A1'),

('B02', 'The 48 Laws of Power', 'Robert Greene', '9780061122415', 'Self help', 1998, 'Borrowed', 'B2'), ('B03', 'Harry Potter', 'J. K. Rowling', '9781250301697', 'Children literature', 1997, 'Available', 'C3'),

('B04', 'Little Women', 'Louisa May Alcott', '9781455586691', 'Fiction', 1868, 'Borrowed', 'A2'),

('B05', 'Becoming', 'Michelle Obama', '9781524763138', 'Biography', 2018, 'Available', 'D1'),

('B06', 'Rich Dad Poor Dad', 'Robert Kiyosaki', '9781455582015', 'Self help', 1997, 'Borrowed', 'E2');

SELECT * FROM Book;

Book_ID	Title	Author	ISBN	Genre	Publication_Year	BStatus	Shelf_Location
B01	Pride and Prejudice	Jane Austen	9780735211292	Fiction	1813	Available	A1
B02	The 48 Laws of Power	Robert Greene	9780061122415	Self help	1998	Borrowed	B2
B03	Harry Potter	J. K. Rowling	9781250301697	Children literature	1997	Available	C3
B04	Little Women	Louisa May Alcott	9781455586691	Fiction	1868	Borrowed	A2
B05	Becoming	Michelle Obama	9781524763138	Biography	2018	Available	D1
B06	Rich Dad Poor Dad	Robert Kiyosaki	9781455582015	Self help	1997	Borrowed	E2

2-INSERT INTO Member VALUES

('M01', 'Deemah', '0501111111', 'Deemah@email.com', '2024-01-10', '2026-01-10', 3, 'Active'),

('M02', 'Bazah', '0502222222', 'Bazah@email.com', '2024-05-22', '2026-05-22', 3, 'Active'),

('M03', 'Nourah', '0503333333', 'Nourah@email.com', '2024-03-15', '2026-03-15', 3, 'Active'),

('M04', 'Aroob', '0504444444', 'Aroob@email.com', '2024-06-01', '2026-06-01', 3, 'Active'),

('M05', 'Sara', '0505555555', 'Sara@email.com', '2023-05-02', '2025-05-02', 3, 'Inactive');

SELECT * FROM Member

Member_ID	Name	Phone	Email	Membership_Date	Expiry_Date	Max_Book_Allowed	MStatus
M01	Deemah	0501111111	Deemah@email.com	2024-01-10	2026-01-10	3	Active
M02	Bazah	0502222222	Bazah@email.com	2024-05-22	2026-05-22	3	Active
M03	Nourah	0503333333	Nourah@email.com	2024-03-15	2026-03-15	3	Active
M04	Aroob	0504444444	Aroob@email.com	2024-06-01	2026-06-01	3	Active
M05	Sara	0505555555	Sara@email.com	2023-05-02	2025-05-02	3	Inactive

3-INSERT INTO Employee VALUES
('E01', 'Reem', 'Librarian', '0561111111', 'Reem@library.com'),

('E02', 'Faisal', 'Assistant', '0562222222', 'Faisal@library.com'),

('E03', 'Abdullah', 'Librarian', '0563333333', 'Abdullah@library.com');

SELECT * FROM Employee;

Employee_ID	Name	Position	Phone	Email
E01	Reem	Librarian	0561111111	Reem@library.com
E02	Faisal	Assistant	0562222222	Faisal@library.com
E03	Abdullah	Librarian	0563333333	Abdullah@library.com

4-INSERT INTO Borrow VALUES
('BR01','B02','M01','E01','2025-10-28', NULL, '2025-11-11','Borrowed', 0.00),

('BR02','B03','M01','E01','2025-10-24', NULL, '2025-11-6','Borrowed', 0.00),

('BR03','B04','M01','E02','2025-10-22', NULL, '2025-11-4','Borrowed', 0.00),

('BR04','B04','M02','E02','2025-10-26', NULL, '2025-11-09','Borrowed', 0.00),

('BR05','B06','M03','E03','2025-10-15', NULL, '2025-10-29','Overdue', 14.00),

('BR06','B01','M04','E02','2025-08-10','2025-08-20','2025-08-25','Returned', 0.00),

('BR07','B03','M02','E01','2025-09-05','2025-09-12','2025-09-15','Returned', 0.00),

('BR08','B01','M01','E01','2025-09-20','2025-10-05','2025-09-27','Returned',16.00),

('BR09','B03','M03','E01','2025-07-01','2025-07-20','2025-07-10','Returned',20.00),

('BR10','B02','M05','E02','2025-04-10','2025-04-20','2025-04-24','Returned', 0.00);

SELECT * FROM Borrow;

Borrow_ID	Book_ID	Member_ID	Employee_ID	Borrow_Date	Actual_Return_Date	Due_Date	BRStatus	Fine
BR01	B02	M01	E01	2025-10-28	null	2025-11-11	Borrowed	0.00
BR02	B03	M01	E01	2025-10-24	null	2025-11-06	Borrowed	0.00
BR03	B04	M01	E02	2025-10-22	null	2025-11-04	Borrowed	0.00
BR04	B04	M02	E02	2025-10-26	null	2025-11-09	Borrowed	0.00
BR05	B06	M03	E03	2025-10-15	null	2025-10-29	Overdue	14.00
BR06	B01	M04	E02	2025-08-10	2025-08-20	2025-08-25	Returned	0.00
BR07	B03	M02	E01	2025-09-05	2025-09-12	2025-09-15	Returned	0.00
BR08	B01	M01	E01	2025-09-20	2025-10-05	2025-09-27	Returned	16.00
BR09	B03	M03	E01	2025-07-01	2025-07-20	2025-07-10	Returned	20.00
BR10	B02	M05	E01	2025-04-10	2025-04-20	2025-04-24	Returned	0.00

5-INSERT INTO Reservation VALUES

('R01','B01','M04','E02','2025-08-05','2025-08-12','Reserved'),

('R02','B03','M02','E01','2025-09-03','2025-09-10','Reserved'),

('R03','B02','M04','E03','2025-10-31','2025-11-05','Pending'),

('R04','B04','M03','E02','2025-10-27','2025-11-03','Pending'),

('R05','B03','M01','E01','2025-07-15','2025-07-22','Cancelled'),

('R06','B05','M05','E01','2025-04-05','2025-04-12','Reserved');

SELECT * FROM Reservation;

Reservation_ID	Book_ID	Member_ID	Employee_ID	Reservation_Date	Expiry_Date	RStatus
R01	B01	M04	E02	2025-08-05	2025-08-12	Reserved
R02	B03	M02	E01	2025-09-03	2025-09-10	Reserved
R03	B02	M04	E03	2025-10-31	2025-11-05	Pending
R04	B04	M03	E02	2025-10-27	2025-11-03	Pending
R05	B03	M01	E01	2025-07-15	2025-07-22	Cancelled
R06	B05	M05	E01	2025-04-05	2025-04-12	Reserved

Data Update/Deletion:

-Update/delete membership details

1-UPDATE Member

SET Expiry_Date = '2027-05-22',

MStatus = 'Active'

WHERE Member_ID = 'M02';

2-DELETE FROM Member

WHERE Member_ID = 'M05';

Member_ID	Name	Phone	Email	Membership_Date	Expiry_Date	Max_Book_Allowed	MStatus
M01	Deemah	0501111111	Deemah@email.com	2024-01-10	2026-01-10	3	Active
M02	Bazah	0502222222	Bazah@email.com	2024-05-22	2027-05-22	3	Active
M03	Nourah	0503333333	Nourah@email.com	2024-03-15	2026-03-15	3	Active
M04	Aroob	0504444444	Aroob@email.com	2024-06-01	2026-06-01	3	Active

-Update book shelf location

1-UPDATE Book

SET Shelf_Location = 'C5'

WHERE Book_ID = 'B03';

Book_ID	Title	Author	ISBN	Genre	Publication_Year	BStatus	Shelf_Location
B01	Pride and Prejudice	Jane Austen	9780735211292	Fiction	1813	Available	A1
B02	The 48 Laws of Power	Robert Greene	9780061122415	Self help	1998	Borrowed	B2
B03	Harry Potter	J. K. Rowling	9781250301697	Children literature	1997	Available	C5
B04	Little Women	Louisa May Alcott	9781455586691	Fiction	1868	Borrowed	A2
B05	Becoming	Michelle Obama	9781524763138	Biography	2018	Available	D1
B06	Rich Dad Poor Dad	Robert Kiyosaki	9781455582015	Self help	1997	Borrowed	E2

-Update/delete contact information for a member

UPDATE Member

SET Phone = '0599999999',

Email = 'Nourah_2005@email.com'

WHERE Member_ID = 'M03'

Member_ID	Name	Phone	Email	Membership_Date	Expiry_Date	Max_Book_Allowed	MStatus
M01	Deemah	0501111111	Deemah@email.com	2024-01-10	2026-01-10	3	Active
M02	Bazah	0502222222	Bazah@email.com	2024-05-22	2027-05-22	3	Active
M03	Nourah	0599999999	Nourah_2005@email.com	2024-03-15	2026-03-15	3	Active
M04	Aroob	0504444444	Aroob@email.com	2024-06-01	2026-06-01	3	Active

Data Queries:

1-List all books currently borrowed by a specific member. (Member=M01)

```
SELECT Book.Title, Borrow.Borrow_Date, Borrow.Due_Date

FROM Borrow

JOIN Book ON Book.Book_ID = Borrow.Book_ID

WHERE Borrow.Member_ID = 'M01'

    AND Borrow.Actual_Return_Date IS NULL;
```

Title	Borrow_Date	Due_Date
The 48 Laws of Power	2025-10-28	2025-11-11
Harry Potter	2025-10-24	2025-11-06
Little Women	2025-10-22	2025-11-04

2-List the top 3 most borrowed books.

```
SELECT Book.Title, COUNT(*) AS Borrow_Count
FROM Borrow
JOIN Book ON Book.Book_ID = Borrow.Book_ID
GROUP BY Book.Book_ID, Book.Title
ORDER BY Borrow_Count DESC
```

```
LIMIT 3;
```

Title	Borrow_Count
Harry Potter	3
Pride and Prejudice	2
Little Women	2

3-List reservations made within the last 7 days.

```
SELECT Reservation.Reservation_ID, Member.Name, Book.Title,
Reservation.Reservation_Date
FROM Reservation
JOIN Member ON Member.Member_ID = Reservation.Member_ID
JOIN Book   ON Book.Book_ID     = Reservation.Book_ID
WHERE Reservation.Reservation_Date >= (CURRENT_DATE - INTERVAL 7 DAY);
```

Reservation_ID	Name	Title	Reservation_Date
R03	Aroob	The 48 Laws of Power	2025-10-31

4-List members who haven’t returned their borrowed books yet.

```
SELECT DISTINCT Member.Member_ID, Member.Name

FROM Borrow
```

JOIN Member ON Member.Member_ID = Borrow.Member_ID

WHERE Borrow.Actual_Return_Date IS NULL;

Member_ID	Name
M01	Deemah
M02	Bazah
M03	Nourah

5-List all books that have never been borrowed.

SELECT Book.Book_ID, Book.Title

FROM Book

LEFT JOIN Borrow ON Borrow.Book_ID = Book.Book_ID

WHERE Borrow.Book_ID IS NULL;

Book_ID	Title
B05	Becoming

6-List the top 3 most borrowed genres.

SELECT Book.Genre, COUNT(*) AS Borrow_Count

```
FROM Borrow

JOIN Book ON Book.Book_ID = Borrow.Book_ID

GROUP BY Book.Genre

ORDER BY Borrow_Count DESC

LIMIT 3;
```

Genre	Borrow_Count
Fiction	4
Children literature	3
Self help	2

7-List which employee processed the most borrowings.

```
SELECT Employee.Name, COUNT(*) AS Total_Borrowings
FROM Borrow
JOIN Employee ON Employee.Employee_ID = Borrow.Employee_ID
GROUP BY Employee.Employee_ID, Employee.Name
ORDER BY Total_Borrowings DESC
LIMIT 1;
```

Name	Total_Borrowings
Reem	5

8-Find the sum of all fines collected on a specific date. (10-5,7-20)

```
SELECT SUM(Fine) AS Total_Fines

FROM Borrow

WHERE Actual_Return_Date IN ('2025-10-05', '2025-07-20');
```

Total_Fines
36.00

9-List members who have reached their maximum allowed book limit.

```
SELECT Member.Member_ID, Member.Name,
    COUNT(Borrow.Borrow_ID) AS Current_Borrowed,
    Member.Max_Book_Allowed
FROM Member
JOIN Borrow ON Borrow.Member_ID = Member.Member_ID
WHERE Borrow.Actual_Return_Date IS NULL
GROUP BY Member.Member_ID, Member.Name, Member.Max_Book_Allowed
HAVING COUNT(Borrow.Borrow_ID) >= Member.Max_Book_Allowed;
```

Member_ID	Name	Current_Borrowed	Max_Book_Allowed
M01	Deemah	3	3

10-List members who returned books late.

```
SELECT Member.Name, Book.Title, Borrow.Due_Date, Borrow.Actual_Return_Date
FROM Borrow
```

```
JOIN Member ON Member.Member_ID = Borrow.Member_ID
JOIN Book   ON Book.Book_ID     = Borrow.Book_ID
WHERE Borrow.Actual_Return_Date > Borrow.Due_Date;
```

Name	Title	Due_Date	Actual_Return_Date
Deemah	Pride and Prejudice	2025-09-27	2025-10-05
Nourah	Harry Potter	2025-07-10	2025-07-20

My Role in the Project:

- **Designed the Enhanced ER Diagram (EER)**
- **Mapped entities to a relational schema**
- **Wrote SQL DDL statements**
- **Developed analytical SQL queries**

- **This project was developed as part of a team, with clearly defined individual responsibilities.**